

Haocheng He

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PROFESSIONAL SUMMARY

Industrial Engineering graduate with a Law minor from GDUT, currently pursuing an MSc in Aviation Engineering and Operations Management at PolyU (GPA: 3.95/4.3). Experience spans airport emergency dispatch simulation, manufacturing line optimization, and AI hardware integration, with 3 published invention patent applications.

EDUCATION

The Hong Kong Polytechnic University <i>Master of Science in Aviation Engineering and Operations Management</i> GPA: 3.95 / 4.3 (Current) Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty	Hong Kong SAR, China Sep 2025 – Jan 2027
Guangdong University of Technology <i>Bachelor of Engineering in Industrial Engineering</i> GPA: 3.67 (Top 20%) Thesis: AnyLogic-Based Simulation Optimization of Emergency Vehicle Route Planning in Large Airports Honors: Excellent Graduation Thesis + Innovation Award	Guangzhou, China Sep 2021 – Jun 2025
<i>Bachelor of Laws (Minor Degree in Law)</i> GPA: 3.61 Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management	Sep 2022 – Jun 2025

EXPERIENCE

Assistant AI Engineer <i>Shenzhen Wuxian Hongyin Technology Co., Ltd.</i> - Supported module integration for a camera- and Ascend-based piano fingering recognition system (Python/C++); reviewed architecture and helped diagnose issues. - Maintained Ubuntu servers and Huawei switches/routers; also supported administration and technical interviews.	May 2025 – Jun 2026 Shenzhen, China
Research Assistant <i>Guangdong University of Technology (with Guangzhou Baiyun Int’l Airport)</i> - Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios. - Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms. - Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).	Sep 2023 – Jun 2025 Guangzhou, China
Industrial Engineering Intern <i>Midea Group — Kitchen & Water Heater Division</i> - Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps. - Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.	Apr 2024 – May 2024 Foshan, China
Manufacturing Operations Intern <i>Guangxi LiuGong Machinery Co., Ltd.</i> - Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis. - Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Submitted recommendations to the production-line team for evaluation.	Jul 2023 – Aug 2023 Liuzhou, China

PROJECTS

Aircraft Engine Remaining Useful Life Prediction (Course Project) <i>Guangdong University of Technology</i> - Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA. - Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.	Apr 2024 – Jul 2024 Guangzhou
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SKILLS & INTELLECTUAL PROPERTY

Programming: Python, Java, SQL/MySQL, C/STM32, JS/TS ML: LSTM, BiLSTM, Transformer, RF, PCA AI: LLM Workflows, Prompt Eng., Loop Engineering
Simulation: AnyLogic, Plant Sim., OR-Tools, Dijkstra, A*, BFS, GA, ECRS Tools: Ubuntu/Linux, Git, JLC EDA, Oscilloscope, Logic Analyzer
Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)
Patent Applications: MCMC Sampling-Based Motion Control Method for a PI Coating Machine Controller (CN122085794A, 2026); Coating Control Method and System Based on Deep Reinforcement Learning (CN121325552A, 2025); Energy Regulation Method and System for a Household Solar Energy Storage System (CN117767437A, 2024). Inventors include Xianzhong Zhou, Haocheng He et al.
Utility Model: Integrated Equipment for Wire Harness Production (CN221352455U, 2024). Software Copyrights: Financial Data Analysis V1.0 (2024SR0146827); Flood Emergency Dispatch V1.0 (2024SR0549247).